

# Information Security Analyst Track

## **SOC T1**

### **Project**

Building and Operating a Mini  
Security Operations Center (SOC)

**Done by :  
El-hout Team**

**Supervised by :  
ENG/ Marwan EL-Sayyad**

**2026**

# Project Overview

This project focuses on building a centralized **Security Information and Event Management (SIEM)** system designed to monitor, detect, and analyze security events across Linux environments

utilizing the **Elastic Stack**

the system provides a robust framework for real-time log management and performance tracking, ensuring a proactive defense against cyber threats

# Objectives

- Centralized Logging
- Real-time Monitoring
- Threat Detection
- Data Visualization



# MINI SOC

## ELK STACK ARCHITECTURE

### 1. DATA COLLECTION - ENDPOINTS



Windows 11 VM  
Security Logs  
Event IDs:  
4625, 4720, 4688



Winlogbeat



Ubuntu Linux Server  
Syslog & Auditd Logs



Filebeat

TLS

### 2. ELK PROCESSING & STORAGE



Logstash  
(Ingestion)

- Parse
- Filter
- Enrich



Elasticsearch  
(Indexing & Storage)

- Index
- Store
- Search



Indices

- logs-\*
- security-\*
- system-\*

### 3. VISUALIZATION & ALERTING



Kibana  
(Dashboards)

- Search
- Visualize
- Dashboard



Alerts & Detection

- Elastic Detection Rules
- Threshold Alerts
- Anomaly Detection



Email  
Alerts



Slack  
Notifications



Webhook  
/ API



MINI SOC

MONITOR • DETECT • RESPOND

#### LAB ENVIRONMENT

- ELK Stack
- Ubuntu 22.04
- 8 GB RAM
- 3 vCPU

#### Kibana Dashboard

Total Events

125,892

Last 24 hours

#### Event Breakdown



#### Events Over Time



#### Logs Explorer

| Time                    | host.name  | event.category | event.action  | message                        |
|-------------------------|------------|----------------|---------------|--------------------------------|
| May 21, 2024 @ 12:31:45 | wint1-vm   | authentication | login-failure | Login failed for user admin... |
| May 21, 2024 @ 12:31:42 | ubuntu-srv | process        | start         | Process started: /usr/bin/sudo |
| May 21, 2024 @ 12:31:41 | ubuntu-srv | system         | login         | User connected: root           |
| May 21, 2024 @ 12:31:37 | wint1-vm   | process        | start         | New process: powershell.exe    |
| May 21, 2024 @ 12:31:35 | ubuntu-srv | authentication | sudo          | User admin executed sudo       |
| May 21, 2024 @ 12:31:32 | wint1-vm   | network        | connection    | Outbound connection to 8.8.8.8 |

#### Security Alerts

| Time                    | Rule                   | Severity | host.name  | Description                    |
|-------------------------|------------------------|----------|------------|--------------------------------|
| May 21, 2024 @ 12:31:45 | Multiple Failed Logins | High     | wint1-vm   | Multiple failed login attempts |
| May 21, 2024 @ 12:31:32 | Process Execution      | Medium   | wint1-vm   | Powershell executed            |
| May 21, 2024 @ 12:31:21 | Sudo Execution         | Medium   | ubuntu-srv | Sudo command executed          |
| May 21, 2024 @ 12:30:55 | Suspicious Network     | High     | ubuntu-srv | Connection to unusual IP       |
| May 21, 2024 @ 12:30:12 | System Login           | Low      | ubuntu-srv | User root logged in            |

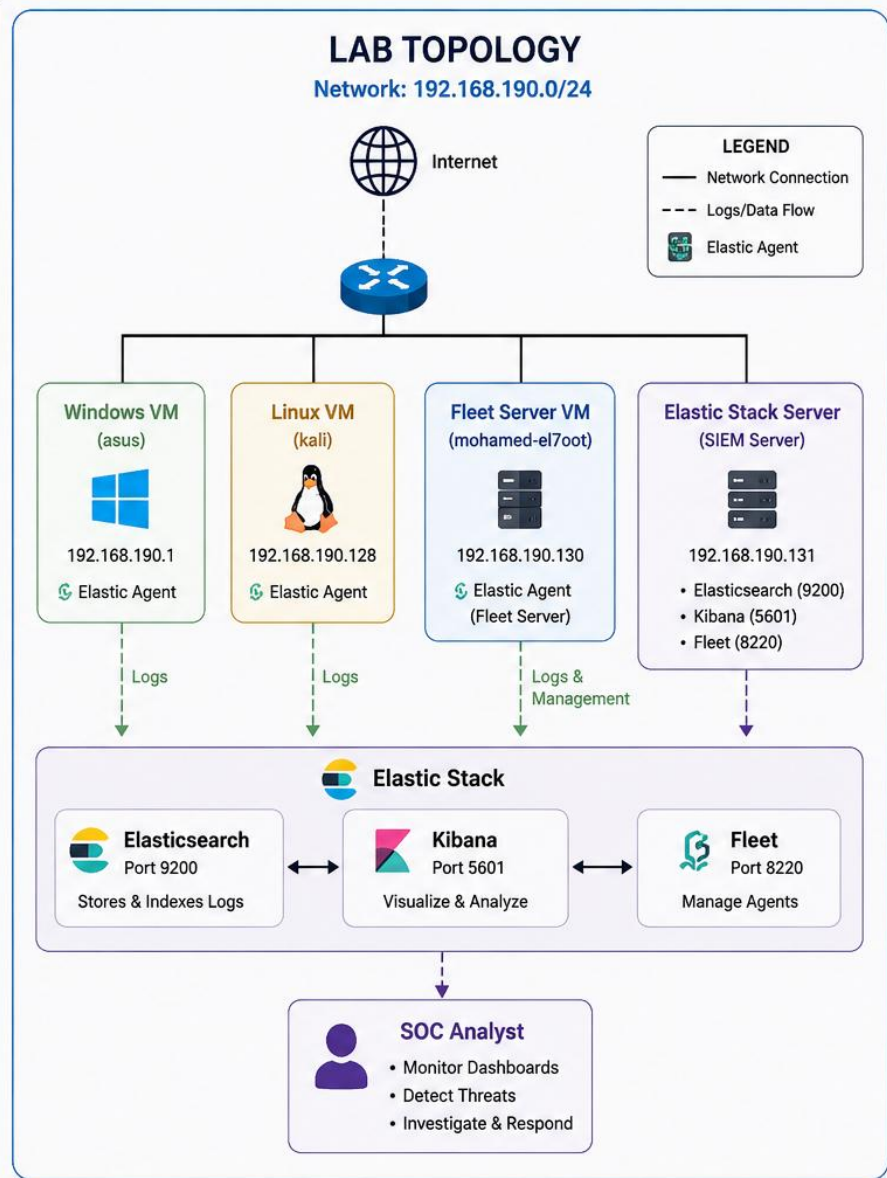
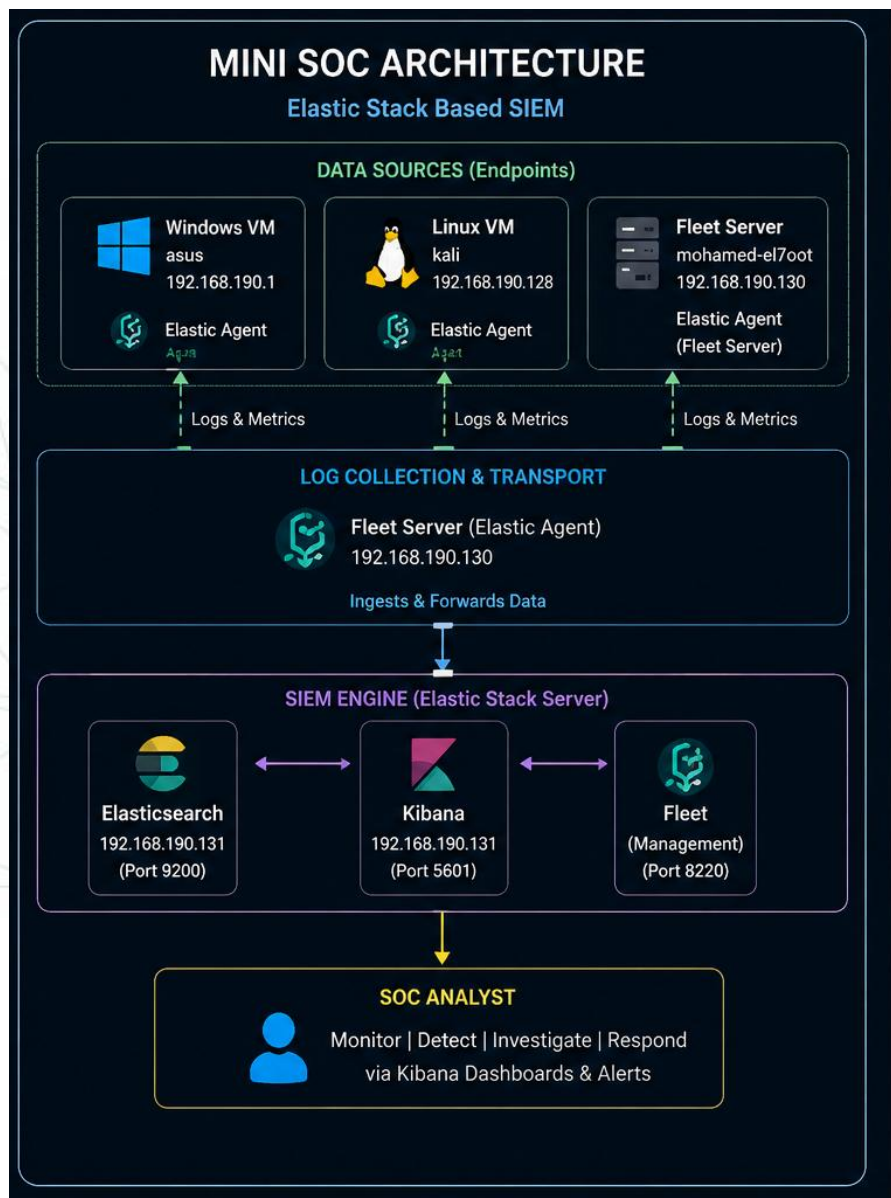
#### Alerts Over Time



#### THREAT HUNTING

- Monitor
- Detect
- Analyze
- Respond

# LAB TOPOLOGY





# Week 1

## SOC Setup & Log Ingestion

### 1-Technical Stack used

- **Operating System:** Ubuntu 22.04 LTS
- **SIEM Engine:** Elasticsearch 8.x
- **Visualization:** Kibana
- **Log Collector:** Elastic Agent

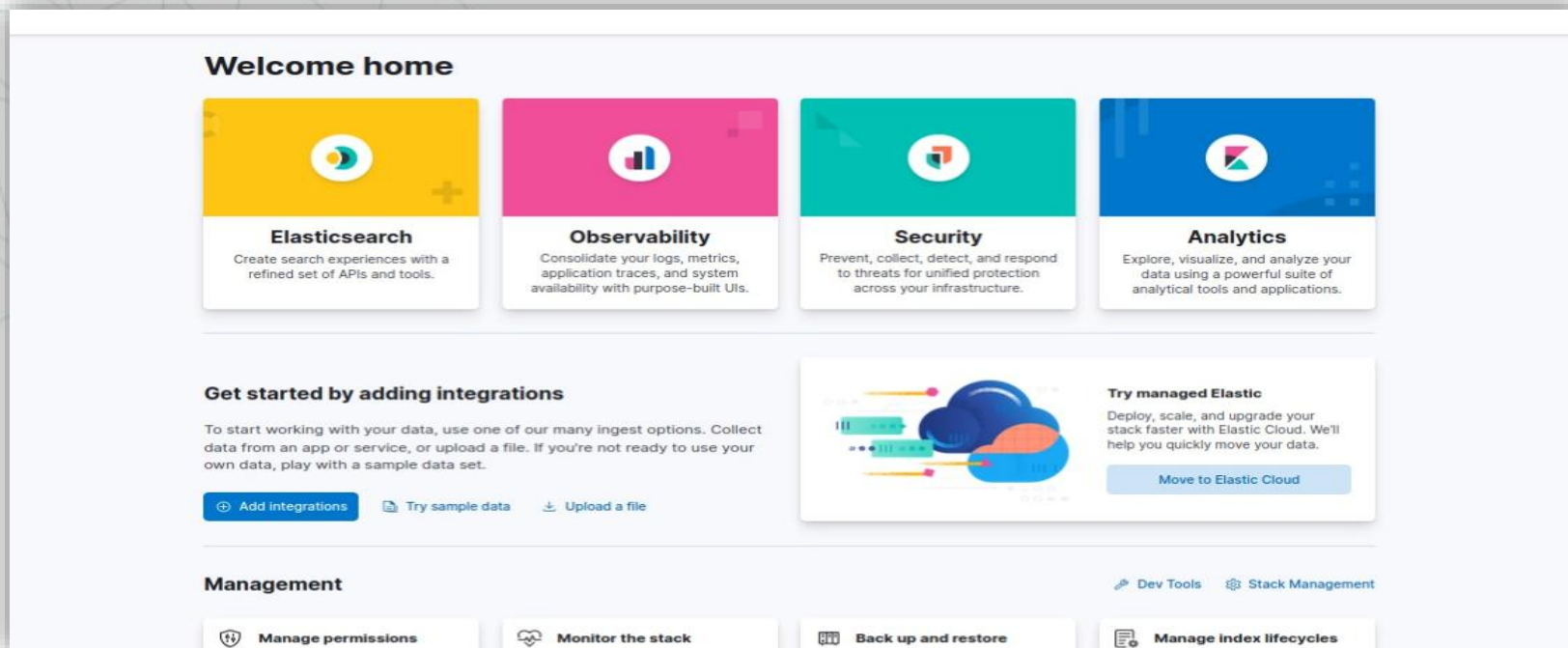
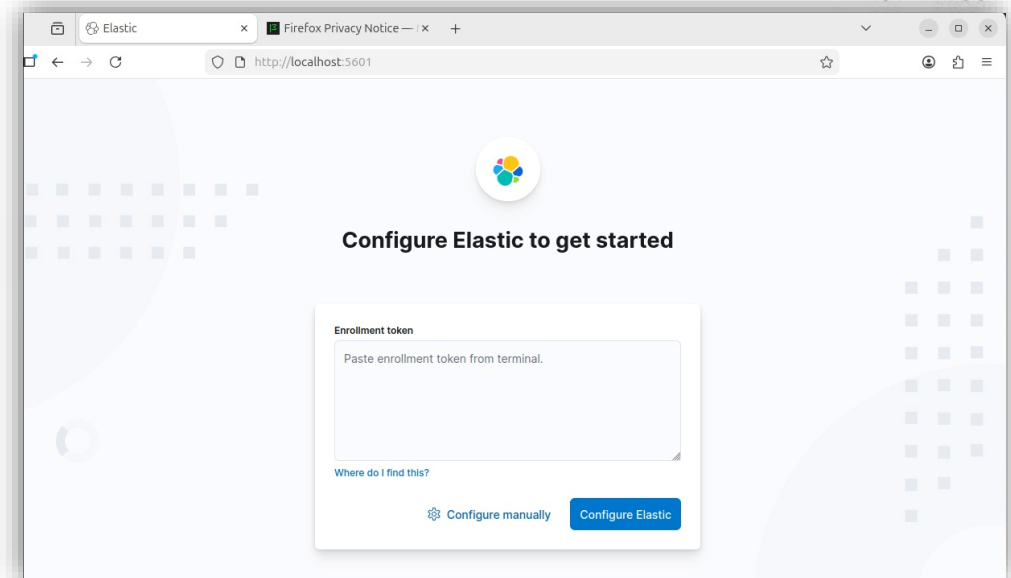
## 2-Implementation Phase

### Step 1: Deploying the Elastic Stack

The ELK environment was set up and the security protocols (Role-based access control) were activated.

**Username** : elastic

**Password** : lEl2kcHY27G\*wB9=mA1T



## Step 2: Elastic Agent Configuration (The Challenge)

- The **Elastic Agent** has been installed.
- We found a problem

SSL Handshake (Error x509)

The problem was resolved by modifying the output configuration of the Elastic Agent .

- Modify the elastic-agent.yml settings file (configuration) to link it to Elasticsearch.

```
ssl.verification_mode: none
```

```
#####  
# Fleet configuration  
#####  
outputs:  
  default:  
    type: elasticsearch  
    hosts: [https://localhost:9200]  
    #api_key: "example-key"  
    username: "elastic"  
    password: "lEI2kcHY27G*WB9=mA1T"  
    preset: balanced  
    ssl.verification_mode: "none"
```



## Step 3: Verification & Testing

- Check that agent is running

```
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$ sudo elastic-agent status
fleet
└─ status: (STOPPED) Not enrolled into Fleet
elastic-agent
└─ status: (HEALTHY) Running
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$
```

- Check that Elastic Agent can read my system log

>> curl -u elastic:lEI2kcHY27G\*wB9=mA1T -k -X GET "https://localhost:9200/\_cat/indices?v"

```
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$ sudo systemctl stop elastic-agent
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$ sudo nano /opt/Elastic/Agent/elastic-agent.yml
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$ sudo systemctl restart elastic-agent
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$ curl -u elastic:lEI2kcHY27G*wB9=mA1T -k -X GET "https://localhost:9200/_cat/indices?v"
health status index uuid pri rep docs.count docs.deleted store.size pri.store.size dataset.size
yellow open .ds-metrics-system.network-default-2026.04.21-000001 x3kE4NI1RGeFQt7LTq2KJw 1 1 26 0 114.6kb 114.6kb 114.6kb
green open .internal.alerts-transform.health.alerts-default-000001 JB407X6yRquUanxrRxtig 1 0 0 0 249b 249b 249b
yellow open .ds-metrics-elastic_agent.filebeat_input-default-2026.04.21-000001 jLsRD0YbQCaV43K4suja-Q 1 1 1 0 29.5kb 29.5kb 29.5kb
green open .internal.alerts-ml.anomaly-detection.alerts-default-000001 u_gFGhBRQZ28SqPgr1o6bg 1 0 0 0 249b 249b 249b
green open .internal.alerts-observability.slo.alerts-default-000001 G3qi7KZ5T9eymFyvmZaDA 1 0 0 0 249b 249b 249b
green open .internal.alerts-default.alerts-default-000001 ksi4DsbsxSwKucZAhFnVl9g 1 0 0 0 249b 249b 249b
green open .internal.alerts-streams.alerts-default-000001 ZtHelp6YTeu8skUyNRC0mg 1 0 0 0 249b 249b 249b
green open .internal.alerts-observability.apm.alerts-default-000001 ha6J- Ez6THGt9jyNqyPJUw 1 0 0 0 249b 249b 249b
green open .internal.alerts-security.attack.discovery.alerts-default-000001 f6neJDb3QQ2jTZSntuyXXA 1 0 0 0 249b 249b 249b
green open .internal.alerts-observability.metrics.alerts-default-000001 SS-E3UVxSJSWHMYdx-Elvg 1 0 0 0 249b 249b 249b
yellow open .ds-logs-elastic_agent-default-2026.04.21-000001 QLQ9iIppSci31oQRKmxB2w 1 1 225 0 57.5kb 57.5kb 57.5kb
green open .internal.alerts-ml.anomaly-detection.health.alerts-default-000001 qWpt95rqT-K7fVEElmN-tA 1 0 0 0 249b 249b 249b
green open .internal.alerts-security.alerts-default-000001 27kvIBa3TNG3sYHVpr1AfQ 1 0 0 0 249b 249b 249b
green open .internal.alerts-stack.alerts-default-000001 Xiz20kPsT0C8szFqTcUcpq 1 0 0 0 249b 249b 249b
yellow open .ds-metrics-elastic_agent.elastic_agent-default-2026.04.21-000001 L_9IqqBqTmelt3WmCCb2uA 1 1 10 0 61.1kb 61.1kb 61.1kb
yellow open .ds-metrics-elastic_agent.metricbeat-default-2026.04.21-000001 CSLlBtChSfwhU4_7TyYnQ 1 1 2 0 52.2kb 52.2kb 52.2kb
yellow open .ds-metrics-system.cpu-default-2026.04.21-000001 HCL36MJdQKAlxQz20Yab0A 1 1 9 0 106kb 106kb 106kb
green open .internal.alerts-observability.logs.alerts-default-000001 jOpRMNAuR4KE7sq-qs-jMw 1 0 0 0 249b 249b 249b
green open .internal.alerts-observability.uptime.alerts-default-000001 mr-X8Ai_Tf0bjg2-19tdIQ 1 0 0 0 249b 249b 249b
yellow open .ds-metrics-system.memory-default-2026.04.21-000001 kUZL8SCTT_qvmyHYtkVRiW 1 1 8 0 101.6kb 101.6kb 101.6kb
yellow open .ds-metrics-elastic_agent.filebeat-default-2026.04.21-000001 bsfEmWQJQ_GuETJ-Uqreaw 1 1 1 0 43.8kb 43.8kb 43.8kb
yellow open .ds-metrics-system.filesystem-default-2026.04.21-000001 1lPNWqrFTR0ccZC1_l1Mw 1 1 176 0 212.5kb 212.5kb 212.5kb
green open .internal.alerts-observability.threshold.alerts-default-000001 bJudfudDSpeflUX5BdYHnw 1 0 0 0 249b 249b 249b
yellow open .ds-logs-elastic_agent.metricbeat-default-2026.04.21-000001 NAAul1_SQ--piHyYvzfT7A 1 1 1018 0 224.8kb 224.8kb 224.8kb
green open .internal.alerts-dataset.quality.alerts-default-000001 BEa1nCddSK6anC4o8eUl3w 1 0 0 0 249b 249b 249b
el7oot@el7oot-VMware-Virtual-Platform:~/elastic-agent-8.19.14-linux-x86_64$
```

# 3-Dashboards & Visualization

# Discover

Home

Analytics

Discover

Dashboards

Canvas

Maps

Machine Learning

Visualize Library

Elasticsearch

Overview

Content

Elasticsearch

Vector Search

Semantic Search

Behavioral Analytics

Observability

Overview

Alerts

Q Search field names

0

Available fields

212

@timestamp

agent.ephemeral\_id

agent.id

agent.name

agent.type

agent.version

beat.stats.beat.host

beat.stats.beat.name

beat.stats.beat.type

beat.stats.beat.uuid

beat.stats.beat.version

beat.stats.cgroup.cpu.id

beat.stats.cgroup.cpu.stats.periods

beat.stats.cgroup.cpu.stats.throttle.dns

beat.stats.cgroup.cpu.stats.throttle.d.periods

beat.stats.cgroup.memory.id

beat.stats.cgroup.memory.mem.usag.e.bytes

beat.stats.cpu.system.ticks

beat.stats.cpu.system.time.ms

Add a field

Auto interval

No breakdown

Apr 21, 2026 @ 17:50:01.382 - Apr 21, 2026 @ 17:51:33.343 (interval: Auto - second)

Documents (283)

Field statistics

Sort fields 1

Q

+

-

📄

@timestamp

Summary

Apr 21, 2026 @ 17:51:33.343

@timestamp Apr 21, 2026 @ 17:51:33.343 agent.ephemeral\_id 1e507293-e3af-41db-aab2-1121213ad2f7 agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e agent.name el7oot-VMware-Virtual-Platform agent.type metricbeat agent.version 8.19.12 data\_stream.dataset system.filesystem data\_stream.namespace default data\_stream.type metrics ecs.version 8.0.0 elastic\_agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e elastic\_agent.snapshot false elastic\_agent.version 8.19.12 event.agent\_id.status auth\_metadata\_missing event.dataset system.filesystem event.duration 1,065,177 event.ingested Apr 21, 2026 @ 17:51:33.343

Apr 21, 2026 @ 17:51:33.343

@timestamp Apr 21, 2026 @ 17:51:33.343 agent.ephemeral\_id 1e507293-e3af-41db-aab2-1121213ad2f7 agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e agent.name el7oot-VMware-Virtual-Platform agent.type metricbeat agent.version 8.19.12 data\_stream.dataset system.filesystem data\_stream.namespace default data\_stream.type metrics ecs.version 8.0.0 elastic\_agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e elastic\_agent.snapshot false elastic\_agent.version 8.19.12 event.agent\_id.status auth\_metadata\_missing event.dataset system.filesystem event.duration 954,089 event.ingested Apr 21, 2026 @ 17:51:33.343

Apr 21, 2026 @ 17:51:33.343

@timestamp Apr 21, 2026 @ 17:51:33.343 agent.ephemeral\_id 1e507293-e3af-41db-aab2-1121213ad2f7 agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e agent.name el7oot-VMware-Virtual-Platform agent.type metricbeat agent.version 8.19.12 data\_stream.dataset system.filesystem data\_stream.namespace default data\_stream.type metrics ecs.version 8.0.0 elastic\_agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e elastic\_agent.snapshot false elastic\_agent.version 8.19.12 event.agent\_id.status auth\_metadata\_missing event.dataset system.filesystem event.duration 603,881 event.ingested Apr 21, 2026 @ 17:51:33.343

Apr 21, 2026 @ 17:51:33.343

@timestamp Apr 21, 2026 @ 17:51:33.343 agent.ephemeral\_id 1e507293-e3af-41db-aab2-1121213ad2f7 agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e agent.name el7oot-VMware-Virtual-Platform agent.type metricbeat agent.version 8.19.12 data\_stream.dataset system.filesystem data\_stream.namespace default data\_stream.type metrics ecs.version 8.0.0 elastic\_agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e elastic\_agent.snapshot false elastic\_agent.version 8.19.12 event.agent\_id.status auth\_metadata\_missing event.dataset system.filesystem event.duration 1,804,281 event.ingested Apr 21, 2026 @ 17:51:33.343

Apr 21, 2026 @ 17:51:33.343

@timestamp Apr 21, 2026 @ 17:51:33.343 agent.ephemeral\_id 1e507293-e3af-41db-aab2-1121213ad2f7 agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e agent.name el7oot-VMware-Virtual-Platform agent.type metricbeat agent.version 8.19.12 data\_stream.dataset system.filesystem data\_stream.namespace default data\_stream.type metrics ecs.version 8.0.0 elastic\_agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e elastic\_agent.snapshot false elastic\_agent.version 8.19.12 event.agent\_id.status auth\_metadata\_missing event.dataset system.filesystem event.duration 1,512,605 event.ingested Apr 21, 2026 @ 17:51:33.343

Apr 21, 2026 @ 17:51:33.343

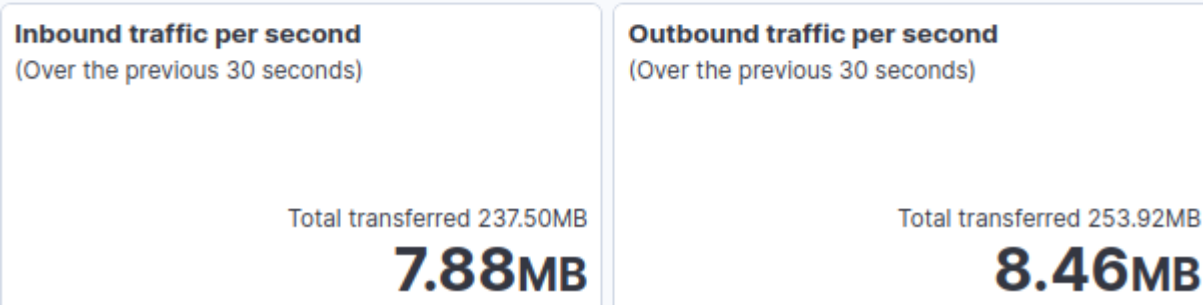
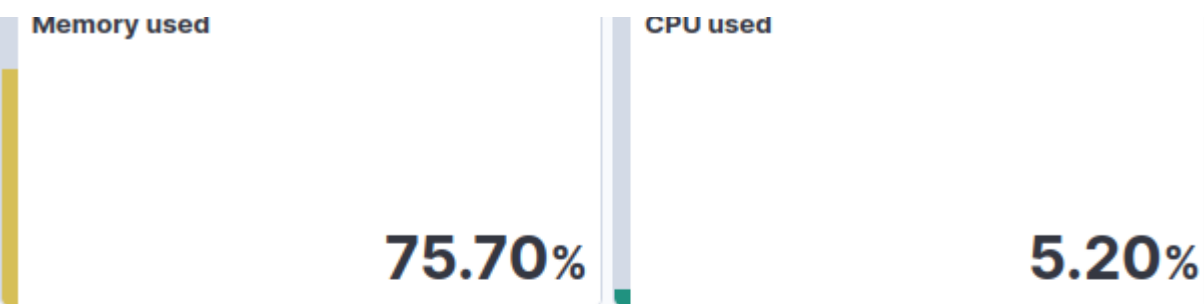
@timestamp Apr 21, 2026 @ 17:51:33.343 agent.ephemeral\_id 1e507293-e3af-41db-aab2-1121213ad2f7 agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e agent.name el7oot-VMware-Virtual-Platform agent.type metricbeat agent.version 8.19.12 data\_stream.dataset system.filesystem data\_stream.namespace default data\_stream.type metrics ecs.version 8.0.0 elastic\_agent.id e376f9a8-8a82-4406-b668-44c87a42ae5e elastic\_agent.snapshot false elastic\_agent.version 8.19.12 event.agent\_id.status auth\_metadata\_missing event.dataset system.filesystem event.duration 670,035 event.ingested Apr 21, 2026 @ 17:51:33.343

Rows per page: 100

1 2 3

## # System overview dashboard

| Host                           | CPU usage | Memory usage |
|--------------------------------|-----------|--------------|
| el7oot-vmware-virtual-platform | 5.20%     | 75.70%       |





# Week 2

# SIEM Deployment and logs check reachability

# Project Overview & Objectives

- **Goal:** Build a complete SIEM environment using open-source Elastic Stack to collect, analyze, and monitor security logs.
- **Scope:**
  - Central Elasticsearch + Kibana
  - Fleet Server to manage agents (Ubuntu server)
  - Windows VM (Asus)
  - Linux VM (kali)
  - Fleet Server VM (mohamed-el7oot)
- **KPIs:**
  - Collect logs from all endpoints
  - Detect security events (e.g., process creation)
  - Centralized visibility in Kibana



# Architecture Overview

## >> Devices used and their IPs

### 1- Elastic Stack Server **192.168.190.131**

- Elasticsearch (port 9200)
- Kibana (port 5601)
- Fleet Server

Ingest Overview Metrics
Agent Info Metrics
Agent activity
Add Fleet Server
Add agent

Status 4
Tags 0
Agent policy 2
Upgrade available

Showing 3 agents
Clear filters
Healthy 3
Unhealthy 0
Updating 0
Offline 0
Inactive 0
Unenrolled 0

| <input type="checkbox"/> | Status  | Host           | Agent policy                  | CPU    | Memory | Last activity  | Version | Actions |
|--------------------------|---------|----------------|-------------------------------|--------|--------|----------------|---------|---------|
| <input type="checkbox"/> | Healthy | asus           | Agent policy 1<br>rev. 1      | N/A    | 302 MB | 24 seconds ago | 8.19.15 | ...     |
| <input type="checkbox"/> | Healthy | kali           | Agent policy 1<br>rev. 1      | 1.43 % | 253 MB | 20 seconds ago | 8.19.15 | ...     |
| <input type="checkbox"/> | Healthy | mohamed-el7oot | Fleet Server Policy<br>rev. 1 | 2.87 % | 281 MB | 14 seconds ago | 8.19.15 | ...     |

# Architecture Overview

2- Linux **192.168.190.128**

attacker device used to deploy some attacks on devices  
by beast using of vulnerabilities

3- Windows **192.168.190.1**

Employee devices are vulnerable to attacks and need protection

## NOTE :

I got IPs

- using (**ip a**) on command line,
- using (**ipconfig**) on CMD

# Fleet Server

## Fleet server hosts

Specify the URLs that your agents will use to connect to a Fleet Server. If multiple URLs exist, Fleet will show the first provided URL for enrollment purposes. For more information, see the [Fleet and Elastic Agent Guide](#).

| Name   | Host URLs                    | Default | Actions                                                                             |
|--------|------------------------------|---------|-------------------------------------------------------------------------------------|
| server | https://192.168.190.131:8220 | ✓       |  |

[+ Add Fleet Server](#)

## Outputs

Specify where agents will send data.

| Name                                                                                        | Type          | Hosts                        | Status | Default                                                   | Actions                                                                               |
|---------------------------------------------------------------------------------------------|---------------|------------------------------|--------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|
| default  | Elasticsearch | https://192.168.190.131:9200 |        | <div>Agent integrations</div> <div>Agent monitoring</div> |  |

# installation of Elastic Agents

## > successfully installed at windows

```
Elastic Agent will be installed at C:\Program Files\Elastic\Agent and will run as a service. Do you want to continue? [Y/n]:y
[ ==] Service Started [5s] Elastic Agent successfully installed, starting enrollment.
[====] Waiting For Enroll... [6s] {"log.level":"warn","@timestamp":"2026-05-19T14:28:03.066+0300","log.logger":"tls","log.origin":{"function":"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file.line":129},"message":"SSL/TLS verifications disabled.","ecs.version":"1.6.0"}
[ ==] Waiting For Enroll... [6s] {"log.level":"info","@timestamp":"2026-05-19T14:28:03.536+0300","log.origin":{"function":"github.com/elastic/elastic-agent/internal/pkg/agent/cmd.(*enrollCmd).enrollWithBackoff","file.name":"cmd/enroll_cmd.go","file.line":545},"message":"Starting enrollment to URL: https://192.168.190.131:8220/","ecs.version":"1.6.0"}
[ ===] Waiting For Enroll... [6s] {"log.level":"warn","@timestamp":"2026-05-19T14:28:03.784+0300","log.logger":"tls","log.origin":{"function":"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file.line":129},"message":"SSL/TLS verifications disabled.","ecs.version":"1.6.0"}
[ =] Waiting For Enroll... [8s] {"log.level":"info","@timestamp":"2026-05-19T14:28:05.187+0300","log.origin":{"function":"github.com/elastic/elastic-agent/internal/pkg/agent/cmd.(*enrollCmd).daemonReloadWithBackoff","file.name":"cmd/enroll_cmd.go","file.line":508},"message":"Restarting agent daemon, attempt 0","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-05-19T14:28:05.192+0300","log.origin":{"function":"github.com/elastic/elastic-agent/internal/pkg/agent/cmd.(*enrollCmd).Execute","file.name":"cmd/enroll_cmd.go","file.line":318},"message":"Successfully triggered restart on running Elastic Agent.","ecs.version":"1.6.0"}
Successfully enrolled the Elastic Agent.
[ =] Done [8s]
Elastic Agent has been successfully installed.
```

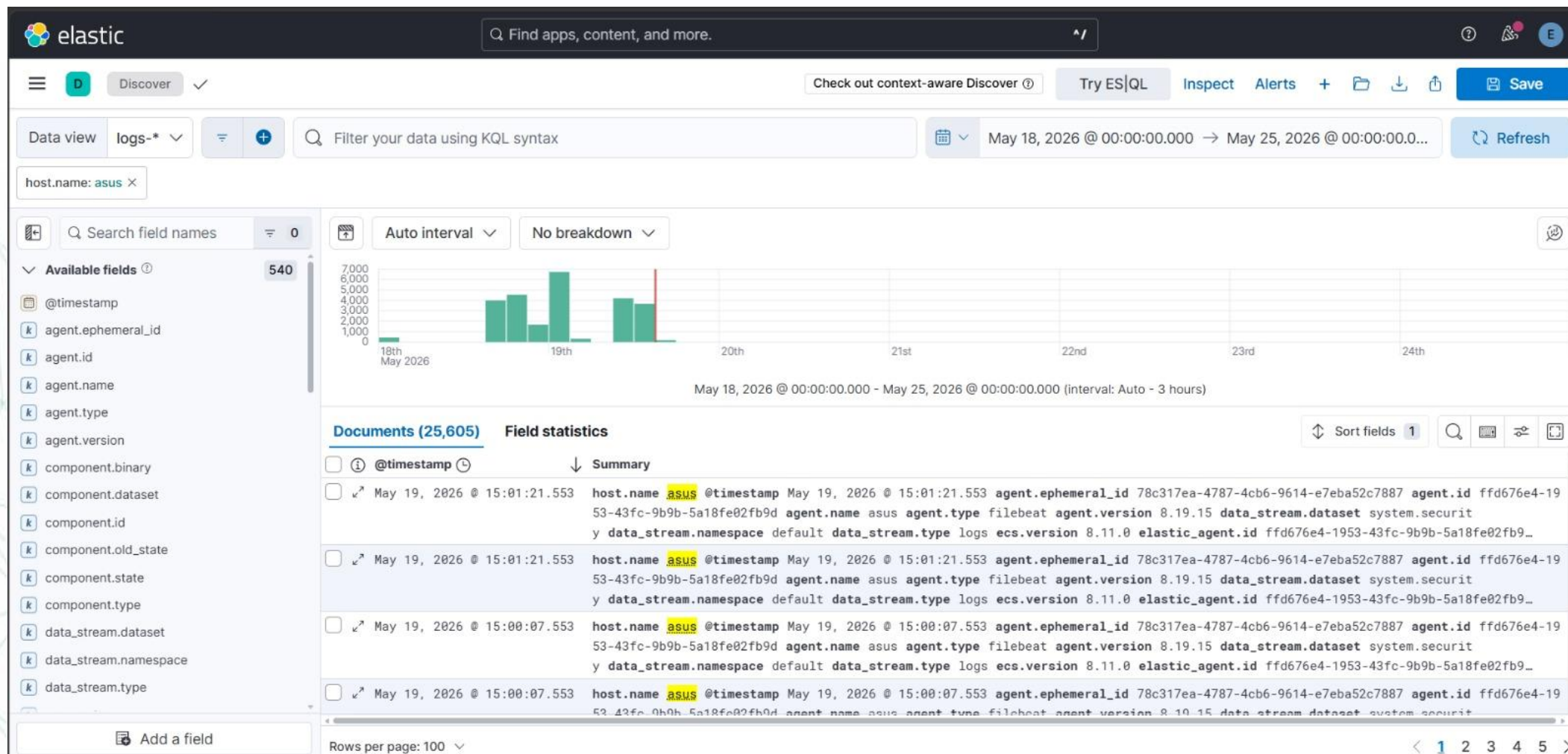
## > successfully installed at Linux

```
Elastic Agent will be installed at /opt/Elastic/Agent and will run as a service. Do you want to continue? [Y/n]:y
[== ] Service Started [8s] Elastic Agent successfully installed, starting enrollment.
[== ] Waiting For Enroll... [8s] {"log.level":"warn","@timestamp":"2026-05-19T06:50:50.380-0400","log.logger":"tls","log.origin":{"function":"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file.line":129},"message":"SSL/TLS verifications disabled.","ecs.version":"1.6.0"}
[== ] Waiting For Enroll... [9s] {"log.level":"info","@timestamp":"2026-05-19T06:50:51.104-0400","log.origin":{"function":"github.com/elastic/elastic-agent/internal/pkg/agent/cmd.(*enrollCmd).enrollWithBackoff","file.name":"cmd/enroll_cmd.go","file.line":545},"message":"Starting enrollment to URL: https://192.168.190.131:8220/","ecs.version":"1.6.0"}
[= ] Waiting For Enroll... [9s] {"log.level":"warn","@timestamp":"2026-05-19T06:50:51.331-0400","log.logger":"tls","log.origin":{"function":"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file.line":129},"message":"SSL/TLS verifications disabled.","ecs.version":"1.6.0"}
[ ] Waiting For Enroll... [11s] {"log.level":"info","@timestamp":"2026-05-19T06:50:53.046-0400","log.origin":{"function":"github.com/elastic/elastic-agent/internal/pkg/agent/cmd.(*enrollCmd).daemonReloadWithBackoff","file.name":"cmd/enroll_cmd.go","file.line":508},"message":"Restarting agent daemon, attempt 0","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-05-19T06:50:53.049-0400","log.origin":{"function":"github.com/elastic/elastic-agent/internal/pkg/agent/cmd.(*enrollCmd).Execute","file.name":"cmd/enroll_cmd.go","file.line":318},"message":"Successfully triggered restart on running Elastic Agent.","ecs.version":"1.6.0"}
Successfully enrolled the Elastic Agent.
[ =] Done [11s]
Elastic Agent has been successfully installed.
```



# Windows Logs Analysis

All logs indexed under "logs-\*" data view



# Linux Logs Analysis (kali)

**elastic** Find apps, content, and more.

Discover ✓ Check out context-aware Discover ⓘ Try ES|QL Inspect Alerts + Save

Data view logs-\* Filter your data using KQL syntax May 18, 2026 @ 00:00:00.000 → May 25, 2026 @ 00:00:00.000 Refresh

host.name: kali

Search field names 0 Auto interval No breakdown

Available fields 540

- @timestamp
- agent.ephemeral\_id
- agent.id
- agent.name
- agent.type
- agent.version
- component.binary
- component.dataset
- component.id
- component.old\_state
- component.state
- component.type
- data\_stream.dataset
- data\_stream.namespace
- data\_stream.type

May 18, 2026 @ 00:00:00.000 - May 25, 2026 @ 00:00:00.000 (interval: Auto - 3 hours)

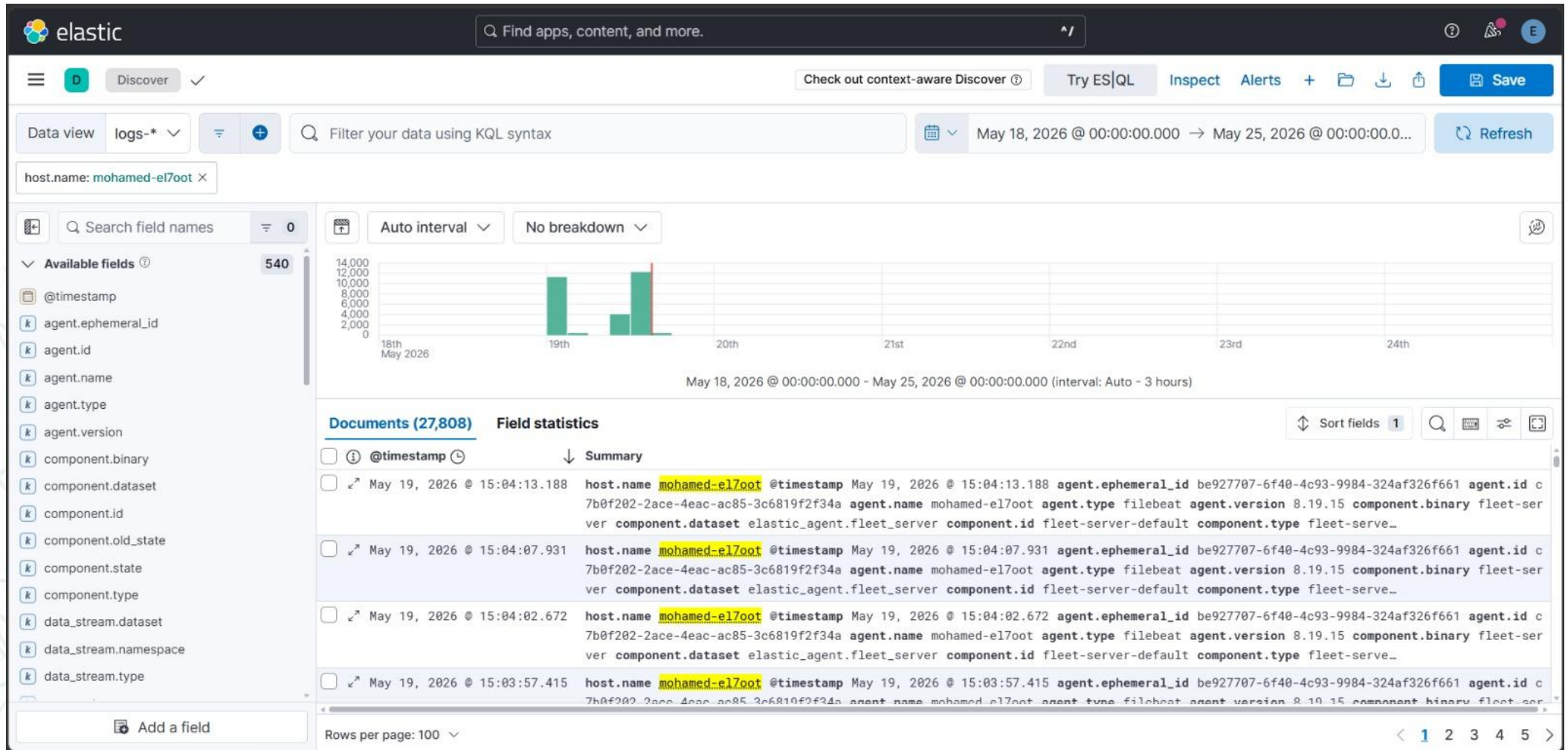
**Documents (192)** Field statistics Sort fields 1

Summary

| @timestamp                  | host.name | agent.ephemeral_id                   | agent.name | agent.type | agent.version | component.binary | component.dataset        | component.id           | component.type   |
|-----------------------------|-----------|--------------------------------------|------------|------------|---------------|------------------|--------------------------|------------------------|------------------|
| May 19, 2026 @ 13:51:10.972 | kali      | 32f5dcf3-793e-40e7-b56a-5f1b5faaa9b1 | kali       | filebeat   | 8.19.15       | metricbeat       | elastic_agent.metricbeat | system/metrics-default | system/metric... |
| May 19, 2026 @ 13:51:10.814 | kali      | 32f5dcf3-793e-40e7-b56a-5f1b5faaa9b1 | kali       | filebeat   | 8.19.15       | metricbeat       | elastic_agent.metricbeat | system/metrics-default | system/metric... |
| May 19, 2026 @ 13:51:10.702 | kali      | 32f5dcf3-793e-40e7-b56a-5f1b5faaa9b1 | kali       | filebeat   | 8.19.15       | metricbeat       | elastic_agent.metricbeat | system/metrics-default | system/metric... |
| May 19, 2026 @ 13:51:00.699 | kali      | 32f5dcf3-793e-40e7-b56a-5f1b5faaa9b1 | kali       | filebeat   | 8.19.15       | metricbeat       | elastic_agent.metricbeat | system/metrics-default | system/metric... |

Rows per page: 100

# Fleet Server Internal Logs





# Security Event Examples

## 1- Event ID 4624 (Successful Logon)

, which is crucial for tracking the attacker's lateral movement or successful access. Additionally,

**2-Event ID 4625 (Failed Logon)** will help us detect and analyze the Brute Force attack attempts.

| Documents (577)          |                                     | Field statistics                                                                                                                                                                                                                                                                                                                                                                                          |  |  | Sort fields 1 | Q |  |  |  |
|--------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------|---|--|--|--|
| <input type="checkbox"/> | <input type="checkbox"/> @timestamp | Summary                                                                                                                                                                                                                                                                                                                                                                                                   |  |  |               |   |  |  |  |
| <input type="checkbox"/> | May 19, 2026 @ 15:01:21.553         | event.code 4624 winlog.event_id 4624 @timestamp May 19, 2026 @ 15:01:21.553 agent.ephemeral_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset syst em.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18f... |  |  |               |   |  |  |  |
| <input type="checkbox"/> | May 19, 2026 @ 15:00:07.553         | event.code 4624 winlog.event_id 4624 @timestamp May 19, 2026 @ 15:00:07.553 agent.ephemeral_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset syst em.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18f... |  |  |               |   |  |  |  |
| <input type="checkbox"/> | May 19, 2026 @ 14:51:58.154         | event.code 4624 winlog.event_id 4624 @timestamp May 19, 2026 @ 14:51:58.154 agent.ephemeral_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset syst em.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18f... |  |  |               |   |  |  |  |
| <input type="checkbox"/> | May 19, 2026 @ 14:46:21.507         | event.code 4624 winlog.event_id 4624 @timestamp May 19, 2026 @ 14:46:21.507 agent.ephemeral_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset syst em.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18f... |  |  |               |   |  |  |  |



**3- Event ID 4688 (Process Creation)** This allows us to monitor any application launch or any command executed via **CMD** or **PowerShell**

**This event ID is not enough if attacker uses Obfuscation (تشفير أو تمويه للأكواد)**

so we use :

**4- Event ID 4104 (Script Block Logging)**

To get full visibility into **PowerShell** executions, which logs the actual content of the scripts and commands being executed, defeating most obfuscation techniques

Documents (27)

Field statistics

Sort fields 1

@timestamp

Summary

↗

May 19, 2026 @ 11:23:13.684

event.code 4688 winlog.event\_id 4688 @timestamp May 19, 2026 @ 11:23:13.684 agent.ephemeral\_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data\_stream.dataset syst em.security data\_stream.namespace default data\_stream.type logs ecs.version 8.11.0 elastic\_agent.id ffd676e4-1953-43fc-9b9b-5a18f...

↗

May 19, 2026 @ 11:23:13.681

event.code 4688 winlog.event\_id 4688 @timestamp May 19, 2026 @ 11:23:13.681 agent.ephemeral\_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data\_stream.dataset syst em.security data\_stream.namespace default data\_stream.type logs ecs.version 8.11.0 elastic\_agent.id ffd676e4-1953-43fc-9b9b-5a18f...

↗

May 19, 2026 @ 11:23:13.649

event.code 4688 winlog.event\_id 4688 @timestamp May 19, 2026 @ 11:23:13.649 agent.ephemeral\_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data\_stream.dataset syst em.security data\_stream.namespace default data\_stream.type logs ecs.version 8.11.0 elastic\_agent.id ffd676e4-1953-43fc-9b9b-5a18f...

↗

May 19, 2026 @ 11:23:13.649

event.code 4688 winlog.event\_id 4688 @timestamp May 19, 2026 @ 11:23:13.649 agent.ephemeral\_id 78c317ea-4787-4cb6-9614-e7eba52c7887 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data\_stream.dataset syst em.security data\_stream.namespace default data\_stream.type logs ecs.version 8.11.0 elastic\_agent.id ffd676e4-1953-43fc-9b9b-5a18f...

Rows per page: 100

<

1

>

**5-Event ID 4720:** A user account was created

**6-Event ID 4732:** A member was added to a security-enabled local group (e.g., Administrators)

| Documents (30)           |                                     | Field statistics                 |                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|--------------------------|-------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <input type="checkbox"/> | <input type="checkbox"/> @timestamp | <input type="checkbox"/> Summary |                                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <input type="checkbox"/> | May 19, 2026 @ 11:24:07.795         | message                          | Cryptographic operation. Subject: Security ID: S-1-5-19 Account Name: LOCAL SERVICE Account Domain: NT AUTHORITY Logon ID: 0x3E5 Cryptographic Parameters: Provider Name: Microsoft Software Key Storage Provider Algorithm Name: RSA Key Name: {6C06566C-D037-4720-9FDE-79E2ABB38F43} Key Type: User key. Cryptographic Operation: Operation: Open Key. Return Code: 0x0 @timestamp May 19, 202... |  |
| <input type="checkbox"/> | May 19, 2026 @ 11:23:22.014         | message                          | Key migration operation. Subject: Security ID: S-1-5-19 Account Name: LOCAL SERVICE Account Domain: NT AUTHORITY Logon ID: 0x3E5 Process Information: Process ID: 13764 Process Creation Time: 2026-05-19T08:23:22.014191400Z Cryptographic Parameters: Provider Name: Microsoft Software Key Storage Provider Algorithm Name: RSA Key Name: {6C06566C-D037-4720-9FDE-79E2ABB38F43} Key Type: Us... |  |
| <input type="checkbox"/> | May 19, 2026 @ 11:23:22.014         | message                          | Key migration operation. Subject: Security ID: S-1-5-19 Account Name: LOCAL SERVICE Account Domain: NT AUTHORITY Logon ID: 0x3E5 Process Information: Process ID: 13764 Process Creation Time: 2026-05-19T08:23:22.014191400Z Cryptographic Parameters: Provider Name: Microsoft Software Key Storage Provider Algorithm Name: RSA Key Name: {6C06566C-D037-4720-9FDE-79E2ABB38F43} Key Type: Us... |  |
| <input type="checkbox"/> | May 19, 2026 @ 11:23:22.014         | message                          | Key migration operation. Subject: Security ID: S-1-5-19 Account Name: LOCAL SERVICE Account Domain: NT AUTHORITY Logon ID: 0x3E5 Process Information: Process ID: 13764 Process Creation Time: 2026-05-19T08:23:22.014191400Z Cryptographic Parameters: Provider Name: Microsoft Software Key Storage Provider Algorithm Name: RSA Key Name: {6C06566C-D037-4720-9FDE-79E2ABB38F43} Key Type: Us... |  |

>> Log filtering allows you to isolate **specific events** or analyze data within **defined time periods**

## Week 3

# Attack & Detection Documentation



# Attack & Detection Documentation

## Use Case 1: Create account attack

- **Tool Used:** Atomic Red Team (ART)/ Command Prompt (Manual Execution)
- **Target Machine (Victim):** Windows 11 VM (Host Name: asus)
- **MITRE ATT&CK Tactic:** Persistence
- **MITRE ATT&CK Technique:** Create Account: Local Account (T1136.001)
- **Simulation Mechanism:** Self-Simulation (The host executes the technique locally to simulate a compromised scenario).

## Attack Scenario Description

This test simulates a post-compromise activity where an attacker creates a new local user account on the system. This ensures the attacker maintains a "backdoor" for persistent access to the victim's environment, allowing them to log back in even if the primary vulnerability or access vector is patched.

## SIEM Rule Deployment

- **Rule Name:** Persistence - Unauthorized Local Admin Account Created
- **Severity:** Medium
- **Detection Logic (Event ID):** 4720



event id: 4720 (إنشاء الحساب).

Jun 17, 2026 @ 01:21:53.901 event.code 4720 winlog.event\_id 4720 @timestamp Jun 17, 2026 @ 01:21:53.901 agent.ephemeral\_id 4f9e76bb-4d58-4b26-a558-9da177eaed4f agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data\_stream.dataset system.security data\_stream.namespace default data\_stream.type logs ecs.version 8.11.0 elastic\_agent.id ffd676e4-1953-43fc-9b9b-5a18f...

4720

Statusopen1SeverityUser

SummaryTrendCountsTreemap

Severity levels

| Levels   | Count |
|----------|-------|
| ● Medium | 152   |

152 alerts

Alerts by name

| Rule name       |
|-----------------|
| create new user |

create new user

Created by: elastic on Jun 20, 2026 @ 01:59:31.248

Updated by: elastic on Jun 20, 2026 @ 01:59:31.248

About

create an unknown user account

Severity

● Medium

Risk score

39

Max alerts per run

100

Definition

Schedule

## Simulate Create unknown user account using CMD

1- @ CMD write : `net user FakeAdmin "Pass1234!" /add`

2- Then click Enter

## Alert Triage :

|               |                                      |
|---------------|--------------------------------------|
| Alert Name    | Create new user                      |
| Severity      | medium                               |
| Host          | Asus                                 |
| MITRE         | T1136.001                            |
| Status        | True positive                        |
| Analyst Notes | New unknown user account are created |



## Use Case 2: Privilege Escalation attack

- **Tool Used:** Command Prompt (Manual Execution)
- **Target Machine (Victim):** Windows 11 VM (Host Name: asus)
- **MITRE ATT&CK Tactic:** Privilege Escalation
- **MITRE ATT&CK Technique:** Account Manipulation (T1098)
- **Simulation Mechanism:** Self-Simulation (The host executes the technique locally to simulate a compromised scenario).

### Attack Scenario Description

This test simulates a post-compromise activity where an attacker adds a previously created or compromised local user account to a high-privileged group, such as the local Administrators group. By doing so, the attacker successfully escalates their privileges, granting them full administrative control over the target machine to disable security controls or execute further malicious actions.

### SIEM Rule Deployment

- **Rule Name:** Privilege Escalation - User Added to Local Administrators Group
- **Severity:** High
- **Detection Logic (Event ID):** 4732

## >>Privilege Escalation

event\_id: 4732 (إضافته للأدمن).

Jun 17, 2026 @ 01:21:54.170 event.code 4732 winlog.event\_id 4732 @timestamp Jun 17, 2026 @ 01:21:54.170 agent.ephemeral\_id 4f9e76bb-4d58-4b26-a558-9da177eaed4f agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data\_stream.dataset system.security data\_stream.namespace default data\_stream.type logs ecs.version 8.11.0 elastic\_agent.id ffd676e4-1953-43fc-9b9b-5a18f...

+

4732

×

📅

Refresh

## Alerts

Assignees

Manage rules

Status open 1

Severity

User

Host

...

Summary

Trend

Counts

Treemap

Severity levels

| Levels | Count |
|--------|-------|
| High   | 10    |

10 alerts

Alerts by name

| Rule name            | Count |
|----------------------|-------|
| Privilege Escalation | 10    |

< 1 >

Top alerts by

host.name

| host.name      | Count |
|----------------|-------|
| asus           | 90%   |
| mohamed-el7oot | 10%   |

## Simulate Privilege Escalation using CMD

1- @ CMD write :

```
net localgroup Administrators FakeAdmin /add
```

2- Then click Enter

# Simulate attack using atomic red team

## Technical Execution Steps on atomic red team

Step 1: Import the Atomic Red Team Module

```
Set-ExecutionPolicy Bypass -Scope Process -Force  
Import-Module "C:\AtomicRedTeam\invoke-atomicredteam\Invoke-AtomicRedTeam.psd1" -Force
```

Step 2: Execute the Specific Attack Test

```
Invoke-AtomicTest T1136.001
```

Depend on **MITRE ATT&CK ID** to simulate a specific attack



## Created a new local user named **NewLocalUser**

```
PS C:\WINDOWS\system32> Set-ExecutionPolicy Bypass -Scope Process -Force; Import-Module "C:\AtomicRedTeam\invoke-atomicredteam\Invoke-AtomicRedTeam.ps1" -Force
PS C:\WINDOWS\system32> Invoke-AtomicTest T1136.001
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1136.001-4 Create a new user in a command prompt
The command completed successfully.
Exit code: 0
Done executing test: T1136.001-4 Create a new user in a command prompt
Executing test: T1136.001-5 Create a new user in PowerShell
Name Enabled Description
-----
T1136.001_PowerShell True
Exit code: 0
Done executing test: T1136.001-5 Create a new user in PowerShell
Executing test: T1136.001-8 Create a new Windows admin user
The command completed successfully.
The command completed successfully.
Exit code: 0
Done executing test: T1136.001-8 Create a new Windows admin user
This script creates a new user, adds it to a local administrator group and then deletes the user.
User 'NewLocalUser' added to the 'Administrators' group.
User name NewLocalUser
Country/region code 000 (System Default)
Comment
User's comment
Account active Yes
Account expires Never
Password last set 6/17/2026 1:21:53 AM
Password expires Never
Password changeable 6/17/2026 1:21:53 AM
Password required Yes
User may change password No
Workstations allowed All
Logon script
User profile
Home directory
Last logon Never
Logon hours allowed All
Local Group Memberships *Administrators
Global Group memberships *None
The command completed successfully.
User 'NewLocalUser' deleted successfully.
Exit code: 0
Done executing test: T1136.001-9 Create a new Windows admin user via .NET
PS C:\WINDOWS\system32>
```

Executing test: T1136.001-9 Create a new Windows admin user via .NET  
User 'NewLocalUser' created successfully.  
Newly Created User Info:  
Full Name NewLocalUser

Successfully added the user to the **Administrators** group  
(Privilege Escalation)

## Alert Triage :

|               |                                                       |
|---------------|-------------------------------------------------------|
| Alert Name    | Privilege Escalation                                  |
| Severity      | High                                                  |
| Host          | Asus                                                  |
| MITRE         | T1098                                                 |
| Status        | True positive                                         |
| Analyst Notes | Unauthorized User Added to Local Administrators Group |

# Attack & Detection Documentation

## Use Case 3: Brute Force attack

- **Tool Used:** Atomic Red Team (ART)
- **Target Machine (Victim):** Windows 11 VM (Host Name: asus)
- **MITRE ATT&CK Tactic:** Credential Access
- **MITRE ATT&CK Technique:** Brute Force ( T1110 )
- **Simulation Mechanism:** Self-Simulation (The host executes the technique locally to simulate a compromised scenario).

## Attack Scenario Description

This test simulates an authentication exploitation tactic where an attacker or an automated script attempts to gain unauthorized access to a system by guessing local account passwords. By executing a rapid looping sequence of incorrect login attempts, the simulation forces the Windows subsystem to trigger defense telemetry, allowing the SOC team to detect malicious credential-hunting patterns before a successful breach occurs.

## SIEM Rule Deployment

- **Rule Name:** Credential Access – Brute Force Detection
- **Severity:** Medium (Can be escalated to High if thresholds are breached within a short timeframe)
- **Detection Logic (Event ID):** 4625

event\_id: 4625 (failed login )

Documents (7) Field statistics

Sort fields 1

Search Filter Settings

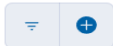
☐ @timestamp

Summary

|                          |                             |                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Jun 20, 2026 @ 00:58:23.757 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:58:23.757 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |
| <input type="checkbox"/> | Jun 20, 2026 @ 00:58:19.228 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:58:19.228 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |
| <input type="checkbox"/> | Jun 20, 2026 @ 00:58:13.407 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:58:13.407 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |
| <input type="checkbox"/> | Jun 20, 2026 @ 00:58:09.371 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:58:09.371 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |
| <input type="checkbox"/> | Jun 20, 2026 @ 00:58:04.618 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:58:04.618 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |
| <input type="checkbox"/> | Jun 20, 2026 @ 00:58:00.875 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:58:00.875 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |
| <input type="checkbox"/> | Jun 20, 2026 @ 00:57:56.678 | event.code 4625 winlog.event_id 4625 @timestamp Jun 20, 2026 @ 00:57:56.678 agent.ephemeral_id d90fab12-1e23-4336-9321-911bb701ab6 2 agent.id ffd676e4-1953-43fc-9b9b-5a18fe02fb9d agent.name asus agent.type filebeat agent.version 8.19.15 data_stream.dataset system.security data_stream.namespace default data_stream.type logs ecs.version 8.11.0 elastic_agent.id ffd676e4-1953-43fc-9b9b-5a18fe0... |

## Alerts

TECHNICAL PREVIEW



Search alerts (e.g. kibana.alert.evaluation.threshold > 7)

Status active

1

Rule

Columns 11 Sort fields 1 1 alert Fields

| Actions | Alert Status | Feature | Last updated |
|---------|--------------|---------|--------------|
| ...     | Active       | Stack   | Jun 20, 2026 |

## Brute Force Detection



1 of 1

Overview

Table

|                      |                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alert Status         | active                                                                                                                                                                |
| Feature              | Stack                                                                                                                                                                 |
| Last updated         | Jun 20, 2026 @ 00:58:29.360                                                                                                                                           |
| Started              | Jun 20, 2026 @ 00:58:29.360                                                                                                                                           |
| Rule category        | Index threshold                                                                                                                                                       |
| Rule                 | Brute Force Detection                                                                                                                                                 |
| Rule tags            | —                                                                                                                                                                     |
| Evaluation values    | —                                                                                                                                                                     |
| Evaluation threshold | —                                                                                                                                                                     |
| Reason               | alert 'Brute Force Detection' is active for group 'all documents': - Value: 6 - Conditions Met: count is greater than 5 over 5m - Timestamp: 2026-06-19T21:58:29.440Z |
| Maintenance windows  | —                                                                                                                                                                     |

## Alerts

Assignees

Manage rules

Status open

1

Severity

User

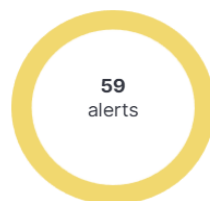
Host

...

Summary Trend Counts Treemap

### Severity levels

| Levels | Count |
|--------|-------|
| Medium | 59    |



### Alerts by name

| Rule name             | Count |
|-----------------------|-------|
| Brute Force Detection | 59    |

1

### Top alerts by

host.name

host.name

asus 100%



## Simulate Brute force attack

1- @ CMD write :

```
runas /user:FakeUser cmd.exe
```

2- Then click Enter

3- Then write any uncorrect password

Repeate this steps 5 times to trigger the alert

## Alert Triage :

|               |                                           |
|---------------|-------------------------------------------|
| Alert Name    | Credential Access – Brute Force Detection |
| Severity      | Medium                                    |
| Host          | Asus                                      |
| MITRE         | T1110                                     |
| Status        | True Positive                             |
| Analyst Notes | Multiple failed login attempts detected   |

# Indicators of Compromise (IOCs)

- **Use Case 1:** Create account attack

Event ID 4720  
Unknown account created

- **Use Case 2:** Privilege Escalation attack

Event ID 4732  
User added to Administrators

- **Use Case 3:** Brute Force attack

Event ID 4625  
Multiple failed logons

# Incident Response Recommendations

Guidance notes and recommended actions to mitigate and resolve the detected attack

- **Create Account**

Disable suspicious account.  
Review local users.  
Audit account creation activities.

- **Privilege Escalation**

Remove unauthorized administrator.  
Review privilege assignments.  
Reset affected credentials.

- **Brute Force**

Enable account lockout policy.  
Change password.  
Enable MFA (**Multi-Factor Authentication**).

## SIEM Rule Deployment & Triggering Workflow:

- **Rule Configuration:** Custom detection rules are built within Kibana to continuously analyze incoming log streams from monitored endpoints.
- **Detection Logic:** Each rule is precisely defined using KQL (Kibana Query Language) to target the specific Windows Event ID that represents a particular attack technique or suspicious activity.
- **Alert Triggering:** When an ingested event matches the predefined Event ID and rule conditions, the SIEM automatically generates and escalates an alert to the security dashboard for analyst review.



## Week 4

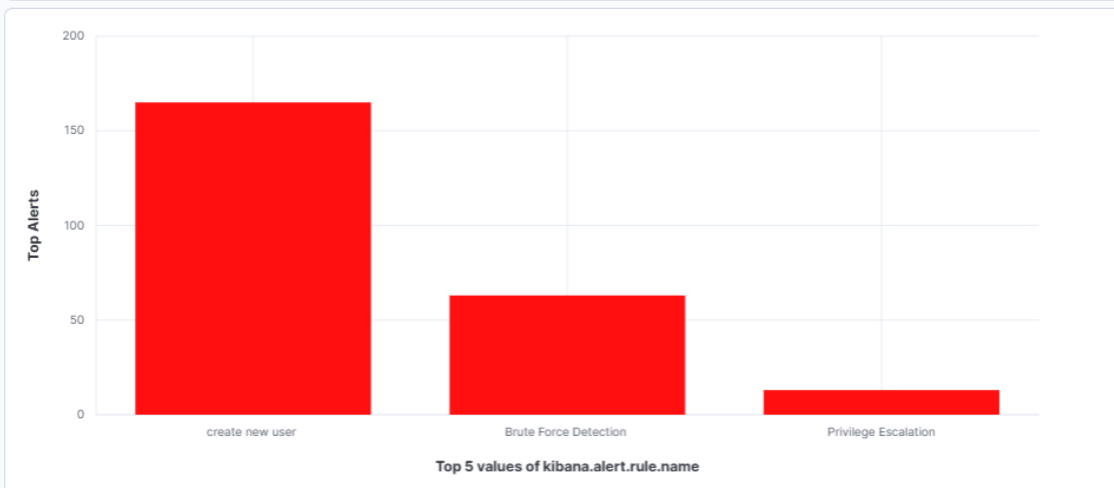
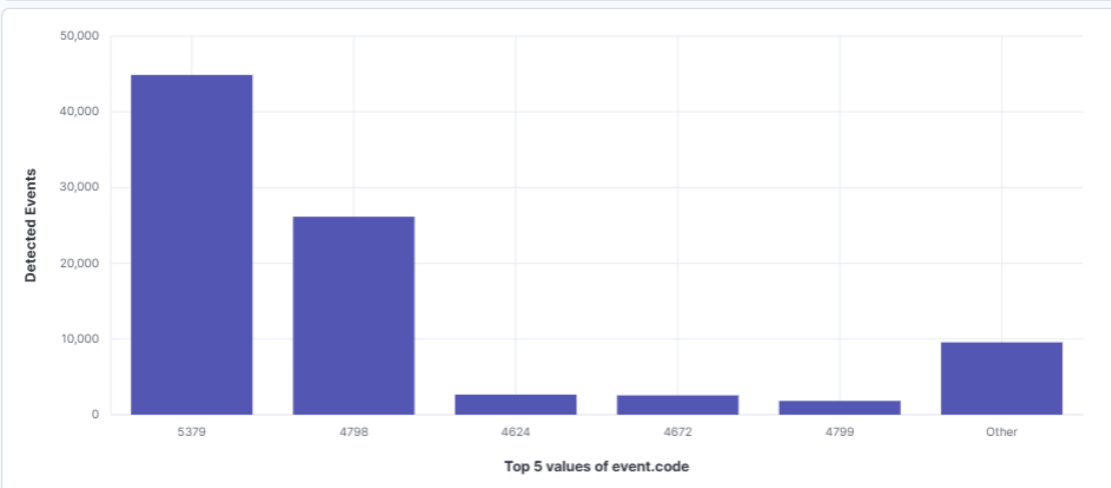
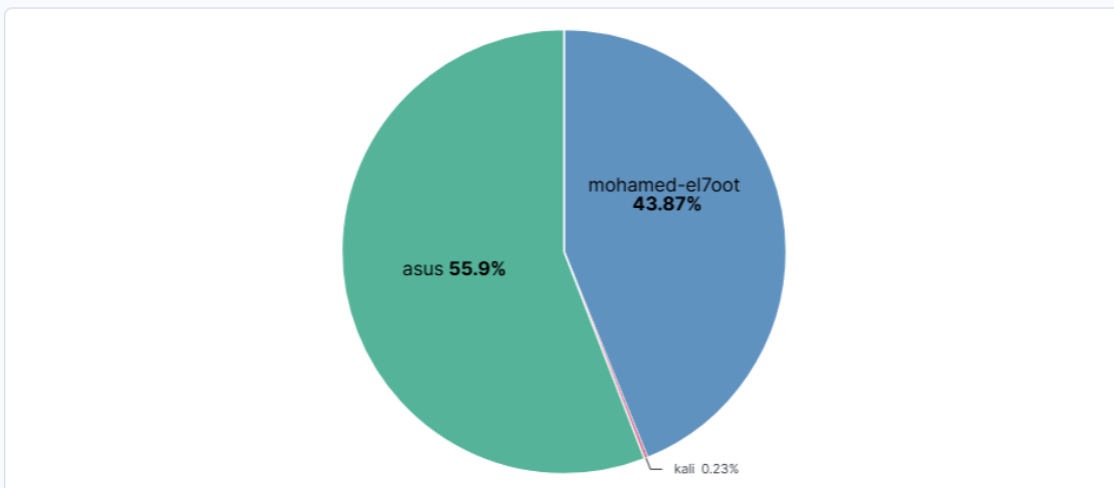
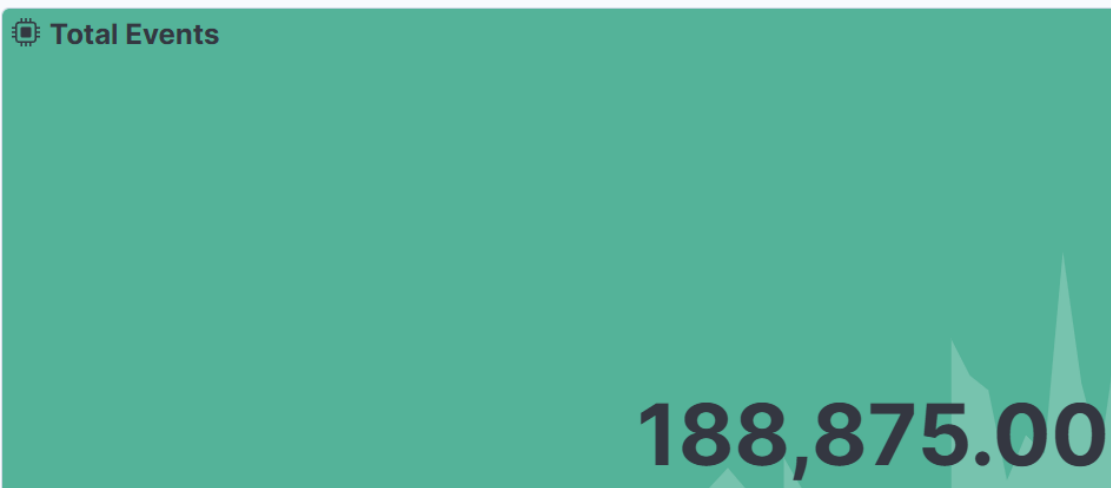
# KPI Dashboard & Security Metrics

# Mini SOC Security Dashboard

Filter your data using KQL syntax

Last 30 days

Refresh



## Total Events

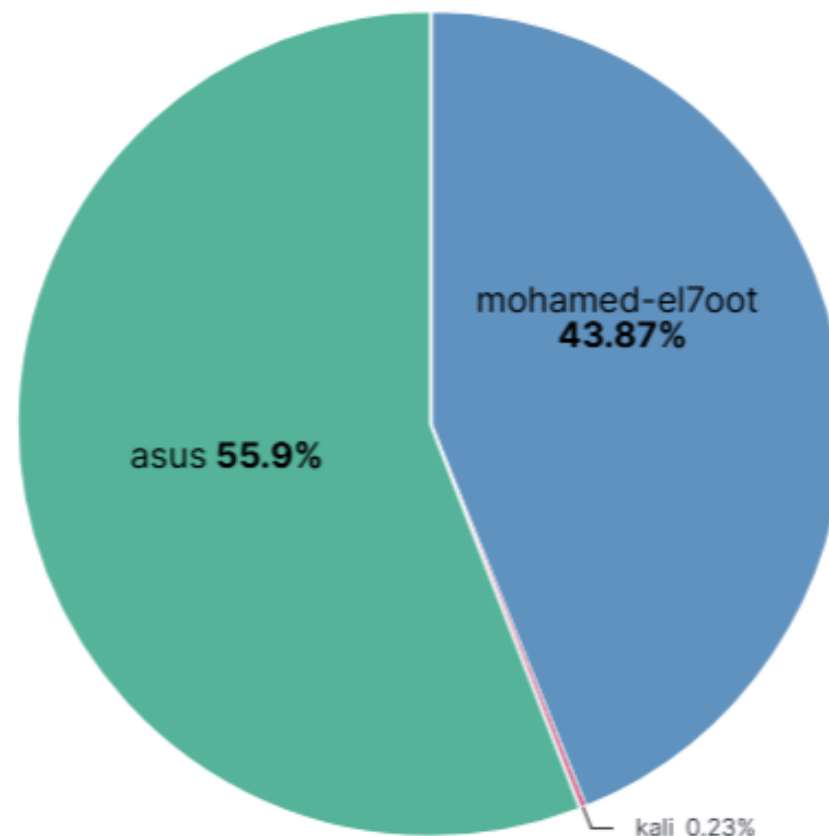
Counts all system events

 Total Events

188,875.00

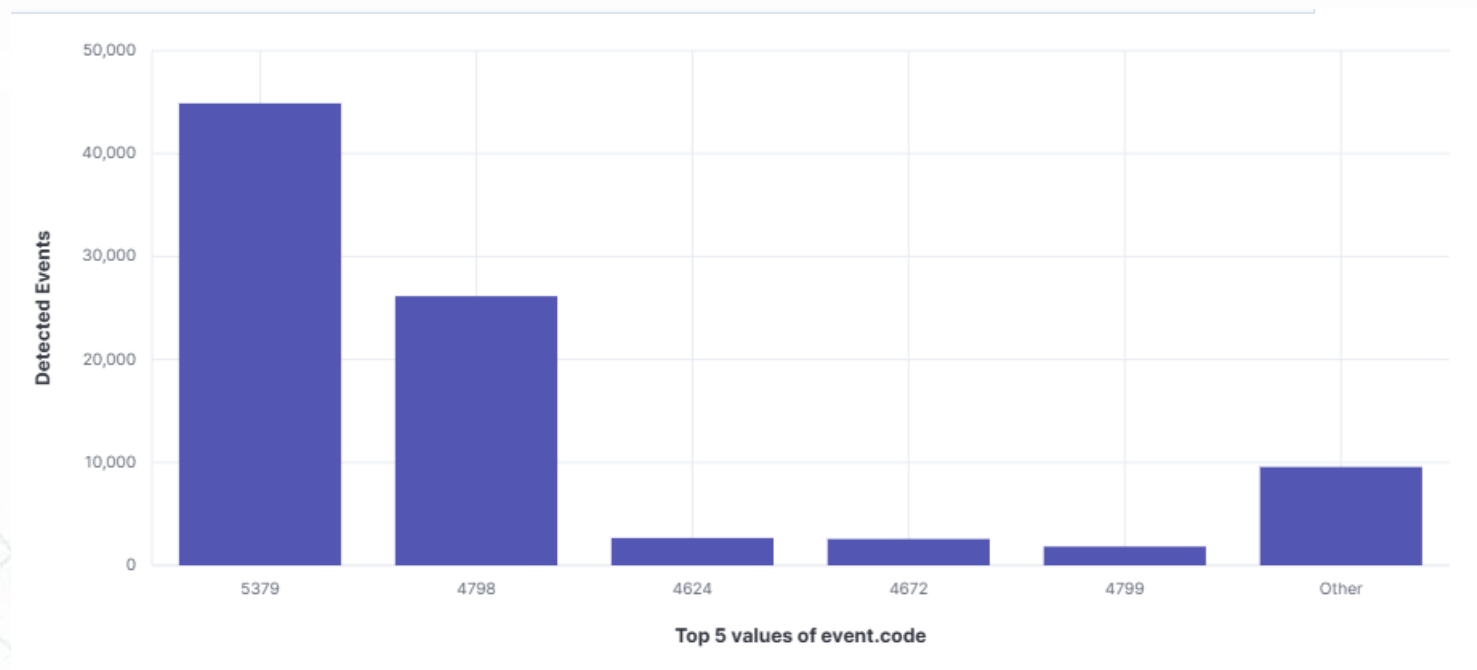
## Hosts Distribution

Display system-monitored endpoints distribution



# Detected Events

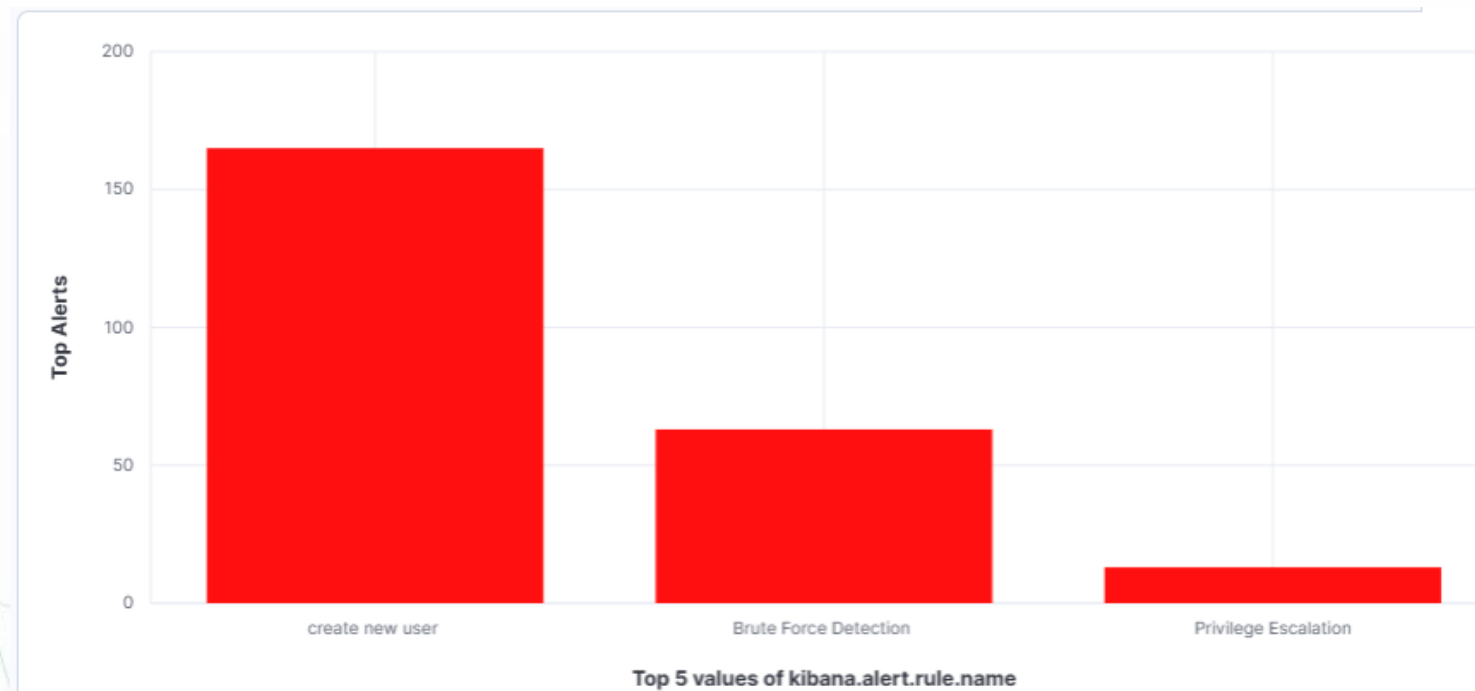
Display system events





# Top Alerts

Display Alerts if there is an attack



## Perform root cause analysis for a selected incident

### Root Cause Analysis

#### Incident

**Brute Force Attack**

#### Detection

Event ID 4625  
Multiple Failed Login Attempts

#### Root Cause

Weak password policy and repeated authentication attempts allowed the attacker perform password guessing attacks.

#### Impact

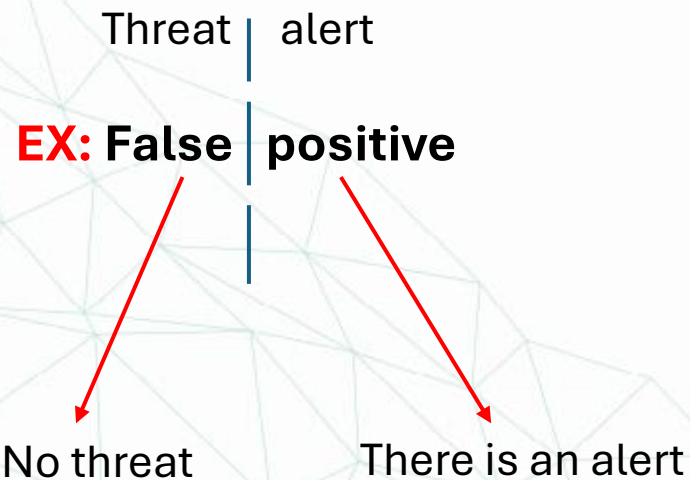
Potential unauthorized access to user accounts.

#### Recommendations

- Enable MFA.
- Implement account lockout policies.
- Use strong passwords.
- Monitor Event ID 4625.

**From the Dashboards section, a new dashboard was created and customized with multiple visualizations to present security KPIs, event statistics, and monitored endpoints. The dashboard serves as a centralized monitoring interface for the SOC team, enabling analysts to quickly detect anomalies, track security incidents, analyze attack patterns, and make informed decisions during incident response activities.**

## Alert triage



**FALSE  
POSITIVE**

Alert triggered,  
no threat



**TRUE  
POSITIVE**

Alert triggered,  
real threat



**FALSE  
NEGATIVE**

Threat not detected,



**TRUE  
NEGATIVE**

Threat not detected,

## Project Summary

- Successfully deployed a Mini SOC using the Elastic Stack.
- Collected and centralized logs from Windows and Linux endpoints.
- Implemented detection rules for multiple attack scenarios.
- Generated and analyzed security alerts.
- Applied MITRE ATT&CK mapping for attack classification.
- Built a KPI dashboard for security monitoring and visibility.



# Thank you